

Changes from `paper3_main_v8` to `v11`

Consolidated record for the co-author. Covers three revision steps:

step	date	nature
v8 → v9	16 Sep 2026	every number regenerated from a single consistent pipeline
v9 → v10	17 Sep 2026	figures consolidated into panels; per-text figures moved to an appendix
v10 → v11	18 Sep 2026	statistics (permutation tests, Spearman), notation collisions, text corrections

Every number is produced by the released code from the tokenised texts with fixed seeds, and is reproduced by `scripts/paper_numbers.py data/corpus`, which writes `results/paper_numbers.md`.

Status column: `done` = already in the `.tex`; `pending` = agreed but not yet applied. Nothing marked `pending` should be taken as final.

0. What matters most, in three points

- The pipeline is reproducible.** The four Pearson correlations regenerate exactly (0.702, 0.295, 0.511, 0.571), verified against `scipy` by an assertion inside `paper_numbers.py`. Everything else in this document rests on that.
- The significance tests changed, and two marks moved.** The parametric t -test on r assumes bivariate normality; the eigenvalue spectrum is strongly skewed, so it was replaced by a permutation test over the pairing of the two spectra. *War and Peace* drops from `***` to `**`, *Ulysses* from `**` to `*`. The substantive conclusions are unchanged.
- One factual error was found and corrected.** The body text claimed PC2 is the burstiest direction of *On the Origin of Species*; the maximum is at PC6 ($B_6 = 1.900$ vs $B_2 = 1.888$). The v10 figure caption already said PC6, so the paper contradicted itself. This also gave us a better argument — see §3.5.

1. v8 → v9 — regenerated numbers

The earlier version mixed results from three separate Colab batches, each of which recomputed PCA and the FGN surrogate independently. The pipeline now computes them once per text, so the burstiness, ACF and DFA tables share the same surrogate (same α_0 , same realisation).

Two consequences worth stating plainly to a referee:

- The FGN surrogate was previously built from a differently tokenised stream (regex $\text{\texttt{(\w+)}}$). DFA exponents in the corpus table shift by up to 0.03 (*Ulysses*: 0.794 $\rightarrow$ 0.763).
- In the previous ACF batch the FGN of every text had been tuned to $\alpha = 0.769$ by a coding slip. The FGN autocorrelation is negligible either way, so the ACF conclusions are unchanged.

1.1 Corpus (status: done)

Alice in Wonderland removed (7,237 filtered tokens — too short for a reliable IET estimate at $q = 0.95$). Corpus is now 16 texts.

quantity	v8	v9+
texts	17	16
raw token range	27,000–565,000	56,000–565,000
filtered token range	12,000–321,000	17,000–236,000
DFA exponent range	[0.61, 0.85]	[0.63, 0.85]

A column α_0 (FGN generator exponent) was added to the corpus table. α_0 is determined only to ± 0.01 by the bisection tolerance and is therefore reported with two decimals throughout.

1.2 Four showcase texts (status: done)

text	α v8 $\rightarrow$ v9	α_0 v8 $\rightarrow$ v9	r v8 $\rightarrow$ v9
War and Peace	0.719 $\rightarrow$ 0.721	0.770 $\rightarrow$ 0.76	0.702 $\rightarrow$ 0.702
Origin of Species	0.769 $\rightarrow$ 0.771	0.804 $\rightarrow$ 0.80	0.294 $\rightarrow$ 0.295
Great Expectations	0.688 $\rightarrow$ 0.689	0.735 $\rightarrow$ 0.74	0.522 $\rightarrow$ 0.511
Ulysses	0.794 $\rightarrow$ 0.763	0.839 $\rightarrow$ 0.80	0.578 $\rightarrow$ 0.571

Mean burstiness over the first 20 components at $q = 0.95$ was previously reported without stating the quantile and with values from a different batch: 1.18 / 1.10 / 1.07 / 1.13 $\rightarrow$ **1.36** (War and Peace), **1.35** (Origin), **1.23** (Ulysses), **1.18** (Great Expectations), against $\bar{B} \simeq 0.97$ –0.98 for both null models in every case.

1.3 Corpus-level burstiness (status: done)

quantity	v8	v9+
$\langle B_k^{\text{orig}} \rangle_{\text{texts}}$	[1.12, 1.37], mean 1.22	[1.16, 1.54], mean 1.26
$\langle B_k^{\text{surr}} \rangle$	[0.972, 0.978]	[0.973, 0.978]
$\langle B_k^{\text{FGN}} \rangle$	[0.970, 0.992]	[0.958, 0.983]
$B^{\text{orig}} > B^{\text{FGN}}$	99.4% of pairs	100% of pairs
$\ B^{\text{FGN}}/B^{\text{surr}} - 1\ < 0.05$	95.9%	95.6%
per-text mean ratio orig/surr	1.248 ± 0.011	1.29 ± 0.03
FGN/surr ratio	$1.0035 \pm 0.0012, t = 2.88, p = 0.011$	$1.001 \pm 0.001, t = 1.1, P = 0.28$

The last row is the most useful change for the argument. In v8 the two null models were *significantly different* from each other, which was awkward to explain away as estimator noise. With the consistent pipeline they are statistically indistinguishable — the cleaner and stronger statement.

The abstract changed accordingly: "exceeds the FGN surrogate in 99.4% of pairs" → "in all pairs".

The excess of the original over the nulls is now described as a $\sim 30\%$ mean excess (was: a consistent $\sim 25\%$ excess). The per-text spread is reported explicitly: [1.17, 1.39] for fifteen texts with *Principia Mathematica* an outlier at 1.73, $\text{sd} = 0.13$.

1.4 ACF and DFA (status: done)

quantity	v8	v9+
GE: $\rho_k(1)$ original range	[0.042, 0.179]	[0.049, 0.208]
GE: i.i.d. band $1.96/\sqrt{M}$	± 0.006	± 0.008
GE: FGN lag-1	< 0.025 , 19/20 inside band	≤ 0.007 , 20/20 inside band
corpus $\bar{\rho}^{\text{orig}}(1)$	0.096 ± 0.044	0.100 ± 0.045
corpus $\bar{\rho}^{\text{FGN}}(1)$	0.003 ± 0.009	0.002 ± 0.005
FGN pairs significant at lag 1	17.2%	10.9%
t -test FGN $\rho(1)$ vs 0	$t = 5.83$	$t = 5.05$
GE: $\bar{\alpha}^{\text{orig}}$	0.637 ± 0.032	0.623 ± 0.034
GE: $\bar{\alpha}^{\text{FGN}}$	0.502 ± 0.015	0.499 ± 0.014
GE: α_k^{orig} range	[0.575, 0.698]	[0.564, 0.693]

A remark about PC1 lying outside the i.i.d. band ($\rho_1^{\text{FGN}}(1) = 0.023$) was **deleted**: with the consistent pipeline the value is 0.004, inside the band.

The residual FGN excess at lag 1 is now quantified rather than asserted: the significant cases concentrate on PC1 (56% of PC1 pairs against 8% for $k \geq 4$), consistent with the known GloVe artefact from rare transliterated proper nouns.

2. v9 → v10 — figure consolidation (status: done)

No numbers changed in this step. Structural only:

- Four per-text burstiness-spectrum figures → one 4-panel figure (`fig:burst_panel`), panels A-D).
- Four per-text quantile-sweep figures → one 4-panel figure (`fig:sweep_panel`).
- The four "semantic poles" figures were **removed**; the same information is in the PC tables (`tab:wrnpc_pc` and siblings), which are more precise and cost less space.
- Per-text eigenvalue spectra and IET distributions moved to a new appendix, *Additional per-text figures* (`app:pertext`), keeping only *War and Peace* in the main text as the worked example.
- Figure paths now follow the corpus filenames (`figures/<stem>/<stem>_NN_<name>.png`), so every figure is regenerable by `scripts/run_text.py`.

- Dropbox references replaced by a pointer to the code repository.

Net effect: roughly fifteen figures down to the panels plus the appendix.

3. v10 → v11 — statistics, notation, corrections

3.1 Notation: $\mathbb{P}$ was overloaded (status: partly done)

$\mathbb{P}$ meant both the **event rate** $p = 1 - q$ (Sec. 2.3, and in the robustness section at $p = 0.01$) and the **p-value** of the correlation and t -tests — both small decimals, a few pages apart. The event rate is structural and appears throughout, so the p-value was renamed $\mathbb{P}$.

Seven occurrences converted (status: done): the four (r, p) pairs in the per-text sections, and three t -test p-values in the corpus sections.

Four occurrences still open (status: **pending**) — they use $\langle \rangle$ and $\langle \rangle$ rather than $\mathbb{P}$, so the first pass missed them:

- table (tab: single_text_summary) footnote: $\$^*p<0.05\$$, $\$^{**}p<0.01\$$, $\$^{***}p<0.001\$$, (no mark: $\$p>0.05\$$)
- $t = 8.7, \text{sqquad } p < 10^{-6}$ (eq. (ratio_corpus))
- $\$t = 5.05\$, \$p < 10^{-6}\$$ (corpus ACF)

3.2 Three further symbol collisions (status: pending)

symbol	conflicting meanings	resolution
α	matching condition in $u_t^{(\alpha)}$ vs DFA exponent (≈ 65 uses)	matching condition $\rightarrow \mathcal{C}$ (6 occurrences)
γ	Zipf exponent $f(r) \sim r^{-\gamma}$ vs diffusion exponent	diffusion exponent $\rightarrow H$, tied explicitly to the DFA α of the stationary series
$\tilde{s}$	retained tokens $(\tilde{s}_1, \dots, \tilde{s}_M)$ vs FGN surrogate word sequence	surrogate $\rightarrow \mathbf{s}^{\text{FGN}}$ (1 occurrence)

H and "Hurst" appear nowhere else in the paper, so this introduces no new collision.

3.3 Significance by permutation, not by the parametric test (status: pending)

The $K = 20$ pairs (λ_k, B_k) are not independent draws: $\{\lambda_k\}$ is a deterministic decreasing sequence and both spectra derive from the same trajectory. The t -test on r is therefore not a valid significance test here. It is replaced by a permutation test over the pairing between the two spectra (10^5 pairings, seed 0), implemented in (scripts/pearsonr_spearman.py) and called from (paper_numbers.py).

text	r	parametric p	permutation P	mark
War and Peace	0.702	0.0006	0.0055	(***) → (**)
Origin of Species	0.295	0.207	0.169	none → none
Great Expectations	0.511	0.021	0.0136	(*) → (*)
Ulysses	0.571	0.009	0.0406	(**) → (*)

The parametric test was anticonservative by up to an order of magnitude, as expected when the predictor is strongly skewed.

Note for the reader of the table: *Ulysses* has a larger r than *Great Expectations* but a larger P . This is not an error — the permutation null depends on the shape of each text's eigenvalue spectrum.

3.4 Spearman added (status: pending)

The eigenvalue spectrum decays steeply, so Pearson is dominated by the leading components. The rank correlation was computed as a robustness check and turns out to carry information of its own:

text	r	ρ	P_ρ
War and Peace	0.702	0.850	$< 10^{-4}$
Origin of Species	0.295	0.460	0.042
Great Expectations	0.511	0.579	0.0085
Ulysses	0.571	0.817	$< 10^{-4}$

$\rho > r$ in all four texts: the association between semantic variance and burstiness is **monotone but not linear**, and Pearson understates it.

($P = 0.0000$) in the regenerated output is the floor of 10^5 permutations ($1/100001$) and is reported as $P < 10^{-4}$, not as zero.

3.5 Text correction: the burstiness maximum of Darwin (status: pending)

The body text of the *Origin of Species* subsection states that PC2 is the burstiest direction. It is not: the maximum is at **PC6**, $B_6 = 1.900$, against $B_2 = 1.888$. The v10 caption of (fig:burst_panel) already gives the maxima correctly as PC3, PC6, PC3, PC2, so the paper currently contradicts itself. The body text is a leftover from v9, when the now-removed per-text figure (fig:darwin_burst) carried the PC2 claim.

This matters beyond the correction. The subsection used the weak correlation as its argument:

The weak eigenvalue-burstiness correlation, $r = 0.295$ ($p = 0.207$), shows especially clearly that the direction of greatest semantic variance need not be the direction of greatest temporal clustering.

That argument no longer survives contact with Spearman: $\rho = 0.460$, $P = 0.042$ is a significant association. The displacement of the maximum to PC6 says the same thing directly, needs no correlation measure at all, and holds in every text (PC3, PC6, PC3, PC2 — never PC1). The subsection is rewritten around that fact, with the two correlations reported as description rather than as the argument.

3.6 Editorial changes (status: done)

- §1.2 retitled *The Altmann et al. mechanism* → *From word-level burstiness to topical organisation*. The mechanism is joint work and the old title over-attributed it; the citation remains in the first sentence.
- "biological variation" → "geographical distribution" in the list of example topics, so the examples match the semantic poles actually found (Darwin's PC2 is geographical).
- $K = 20$ stated once, where the burstiness spectrum is defined, with a note that higher-order components add progressively less variance and leave the qualitative picture unchanged.
- Redundant restatements of "PCA maximises variance, not burstiness" and of the word-shuffled null's properties compressed.

4. Open items

1. Apply the four remaining $\textcircled{p} \rightarrow \textcircled{P}$ (§3.1) and the three symbol renames (§3.2).
 2. Table `tab:single_text_summary`: new marks on r , new ρ column, caption naming the permutation test.
 3. Rewrite of the *Origin of Species* subsection (§3.5).
 4. Check the other three per-text subsections for the same class of error — any claim that PC1, or the wrong PC, is the burstiest.
 5. `paper_numbers.py` sections 2–5 still carry hardcoded "(old ...)" strings from v8. The `.tex` itself is correct and in sync; only the script's comparison column is stale, so its output for those sections currently flags differences that do not exist.
 6. Decide whether ρ goes in the table as its own column or is mentioned once in the methodology. Current recommendation: the column, since ρ says something r does not.
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5. Reproducing everything

bash

```
python scripts/run_corpus.py data/corpus          # corpus_*.csv
python scripts/paper_numbers.py data/corpus       # results/paper_numbers.md
python scripts/run_text.py data/corpus/<text>     # per-text figures
```

`paper_numbers.py` asserts that the permutation implementation reproduces `scipy.stats.pearsonr` to 10^{-9} on every text, so a silent divergence between the two code paths cannot go unnoticed.

Permutation parameters are fixed: 10^5 pairings, seed 0. Both belong in the methodology section alongside the test itself.